

# Jongmin Moon

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## RESEARCH KEYWORDS

Computational Neuroscience · Systems Neuroscience · Vision · Decision Making · Neural Coding

## APPOINTMENTS

|                |                                                                                                                                                                                                            |
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| 2024 – present | <b>Postdoctoral Fellow</b><br>Center for Perceptual Systems<br>The University of Texas at Austin, Austin, TX, United States <ul style="list-style-type: none"> <li>• Advisor: Robbe L. T. Goris</li> </ul> |
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## EDUCATION

|             |                                                                                                                                                                                                                                                                                                                         |
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| 2018 – 2024 | <b>Ph.D.</b> , Biomedical Engineering<br>Ulsan National Institute of Science and Technology, Ulsan, South Korea <ul style="list-style-type: none"> <li>• Advisor: Oh-Sang Kwon</li> <li>• Thesis title: <i>Common computational principles underlying biases in visual perception, memory and confidence</i></li> </ul> |
| 2022        | Computational Neuroscience: Vision<br>Cold Spring Harbor Laboratory, Lloyd Harbor, NY, United States                                                                                                                                                                                                                    |
| 2012 – 2018 | <b>B.S.</b> , Human Factors Engineering; System Design and Control Engineering<br>Ulsan National Institute of Science and Technology, Ulsan, South Korea <ul style="list-style-type: none"> <li>• Compulsory military service (2013 – 2015)</li> </ul>                                                                  |

## PUBLICATIONS

### Peer-Reviewed Journal Articles

1. Saftari, L. N.<sup>#</sup>, **Moon, J.**<sup>#</sup>, & Kwon, O.-S. (2025). Environmental motion presented ahead of self-motion modulates heading direction estimation. *PLOS Computational Biology*, 21(10), e1013571. <https://doi.org/10.1371/journal.pcbi.1013571>
2. **Moon, J.**, Tadin, D.\* & Kwon, O.-S.\* (2023). A key role of orientation in the coding of visual motion direction. *Psychonomic Bulletin & Review*, 30(2), 564–574. <https://doi.org/10.3758/s13423-022-02181-2>
3. **Moon, J.**, & Kwon, O.-S. (2022). Attractive and repulsive effects of sensory history concurrently shape visual perception. *BMC Biology*, 20, 247. <https://doi.org/10.1186/s12915-022-01444-7>
4. **Moon, J.**<sup>#</sup>, Choe, S.<sup>#</sup>, Lee, S., & Kwon, O.-S. (2019). Temporal dynamics of visual attention allocation. *Scientific Reports*, 9, 3664. <https://doi.org/10.1038/s41598-019-40281-7>

### In Preparation

5. **Moon, J.**, Chung, D., Kim, H. J., Sul, S., & Kwon, O.-S. (in preparation). Optimal confidence judgments in perceptual estimations.
6. **Moon, J.**, Yoon, H., & Kwon, O.-S. (in preparation). A behavioral signature of efficient coding in working memory representations.

## PRESENTATIONS

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### Conferences and Workshops

1. Yoon, H., **Moon, J.**, & Kwon, O.-S. (2025, May 16–20). *Attractive serial dependence occurs after motion repulsion*. [Poster presentation]. Annual Meeting of the Vision Sciences Society, St. Pete Beach, FL, United States.
2. **Moon, J.**, Raj G. P., A., & Goris, R. L. T. (2025, April 25–27). *Bayesian confidence paradox*. [Poster presentation]. Workshop on Natural Environments, Tasks, and Intelligence, Austin, TX, United States.
3. **Moon, J.**, Yoon, H., & Kwon, O.-S. (2024, May 17–22). *Probing bidirectional serial dependence in an N-back orientation estimation task*. [Oral presentation]. Annual Meeting of the Vision Sciences Society, St. Pete Beach, FL, United States.
4. **Moon, J.**, Saftari, L. N., & Kwon, O.-S. (2023, August 24–27). *Environmental motion presented ahead of self-motion modulates the heading direction estimation*. [Poster presentation]. Conference on Cognitive Computational Neuroscience, Oxford, United Kingdom.
5. Saftari, L. N., **Moon, J.**, & Kwon, O.-S. (2023, May 19–24). *Synchronicity of visual and vestibular signals modulates the causal inference in heading direction estimation*. [Poster presentation]. Annual Meeting of the Vision Sciences Society, St. Pete Beach, FL, United States.
6. **Moon, J.**, & Kwon, O.-S. (2022, May 13–18). *Confidence in perceptual estimation reflects behavioral variability, but not biases*. [Poster presentation]. Annual Meeting of the Vision Sciences Society, St. Pete Beach, FL, United States.
7. **Moon, J.**, & Kwon, O.-S. (2020, June 19–24). *Motion perception is biased toward the orientation, not the direction, of the preceding motion*. [Poster presentation]. Annual Meeting of the Vision Sciences Society, virtual.
8. **Moon, J.**, & Kwon, O.-S. (2019, May 17–22). *Additivity of attractive and repulsive sequential effects in motion direction estimation*. [Oral presentation]. Annual Meeting of the Vision Sciences Society, St. Pete Beach, FL, United States.
9. Choe, S., **Moon, J.**, & Kwon, O.-S. (2018, May 18–23). *Temporal dynamics of visual attention allocation*. [Poster presentation]. Annual Meeting of the Vision Sciences Society, St. Pete Beach, FL, United States.

### Domestic Conferences

10. Yoon, H., **Moon, J.**, & Kwon, O.-S. (2025, February 12–14). *Attractive serial dependence occurs after motion repulsion*. [Poster presentation]. 62nd Annual Meeting of the Korean Society of Cognitive and Biological Psychology, Gwangju, South Korea.
11. **Moon, J.**, Yoon, H., & Kwon, O.-S. (2024, February 1–3). *Probing bidirectional serial dependence in an N-back orientation estimation task*. [Oral presentation]. 61st Annual Meeting of the Korean Society of Cognitive and Biological Psychology, Seoul, South Korea.
12. **Moon, J.**, Saftari, L. N., & Kwon, O.-S. (2023, February 16–17). *An environmental motion presented ahead of body motion modulates the causal inference in heading direction estimation*. [Oral presentation]. 60th Annual Meeting of the Korean Society of Cognitive and Biological Psychology, Seoul, South Korea.
13. **Moon, J.**, & Kwon, O.-S. (2021, July 15–16). *The effects of behavioral variability and biases on confidence in perceptual estimation*. [Oral presentation]. 58th Annual Meeting of the Korean Society of Cognitive and Biological Psychology, virtual.
14. **Moon, J.**, & Kwon, O.-S. (2020, August 27–28). *A key role of non-directional orientation in the coding of visual motion direction*. [Oral presentation]. 57th Annual Meeting of the Korean Society of Cognitive and Biological Psychology, virtual.
15. **Moon, J.**, & Kwon, O.-S. (2019, February 14–15). *Sequential effects in motion direction estimation*. [Oral presentation]. 56th Annual Meeting of the Korean Society of Cognitive and Biological Psychology, Pyeongchang, South Korea.
16. Choe, S., **Moon, J.**, & Kwon, O.-S. (2018, January 25–26). *Temporal dynamics of visual attention*. [Poster presentation]. 55th Annual Meeting of the Korean Society of Cognitive and Biological Psychology, Suwon, South Korea.

## HONORS AND AWARDS

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|             |                                 |                                                       |
|-------------|---------------------------------|-------------------------------------------------------|
| 2024        | Best Presentation Award         | Korean Society of Cognitive and Biological Psychology |
| 2024        | 2nd Place in Data Challenge     | Korean Society of Cognitive and Biological Psychology |
| 2023        | Best Presentation Award         | Korean Society of Cognitive and Biological Psychology |
| 2022        | CSHL Course Award               | Howard Hughes Medical Institute                       |
| 2021        | Best Presentation Award         | Korean Society of Cognitive and Biological Psychology |
| 2018 – 2024 | UNIST Graduate Scholarship      | Ulsan National Institute of Science and Technology    |
| 2016        | Samsung Dream Class Scholarship | Samsung Welfare Foundation                            |
| 2012        | UNIST Dream Scholarship         | Ulsan National Institute of Science and Technology    |
| 2012 – 2018 | UNIST Undergraduate Scholarship | Ulsan National Institute of Science and Technology    |

## TEACHING

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### Guest Lectures

|      |              |                                                   |
|------|--------------|---------------------------------------------------|
| 2025 | INP Bootcamp | Interdisciplinary Neuroscience Program, UT Austin |
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### Teaching Assistant

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|-------------|---------------------------|---------------------------------------------|
| 2019 – 2023 | Experimental Design       | Department of Biomedical Engineering, UNIST |
| 2019 – 2022 | Sensation and Perception  | Department of Biomedical Engineering, UNIST |
| 2017 – 2018 | Science of Human Behavior | School of Liberal Arts, UNIST               |

### Educational Outreach

|      |                                   |                                            |
|------|-----------------------------------|--------------------------------------------|
| 2016 | World Friends Korea IT (in Ghana) | National Information Society Agency        |
| 2016 | Samsung Dream Class               | Samsung Welfare Foundation                 |
| 2014 | Educational Outreach Program      | ROK Air Force Education & Training Command |

## SERVICE

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### Departmental service

|             |                            |                                          |
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| 2024 – 2026 | CPS Colloquium Coordinator | Center for Perceptual Systems, UT Austin |
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### Ad Hoc Reviewer – Journal

BMC Biology  
eLife  
Journal of Vision  
Nature Communications (co-review)  
Psychological Research  
STAR Protocols

### Ad Hoc Reviewer – Conference

Cognitive Computational Neuroscience

## REFERENCES

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| Robbe L. T. Goris, Ph.D.<br>Associate Professor<br>Center for Perceptual Systems<br>The University of Texas at Austin<br><a href="mailto:robbe.goris@utexas.edu">robbe.goris@utexas.edu</a> | Oh-Sang Kwon, Ph.D.<br>Associate Professor<br>Department of Biomedical Engineering<br>Ulsan National Institute of Science and Technology<br><a href="mailto:oskwon@unist.ac.kr">oskwon@unist.ac.kr</a> |
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